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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-ISE-004

1. Topic (12 words max)

Smart bus route monitoring and revenue management system

2. The research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information System Engineering (ISE)

4. If a continuation of a previous project:

Project ID	Assessment
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's bus transportation sector plays a crucial role in daily mobility for millions of people, including workers, students, and rural communities, who rely on affordable public transportation. However, the system faces persistent regulatory and operational challenges. A central issue is **route noncompliance**, where buses deviate from their permitted routes, causing service gaps, commuter inconvenience and weakened oversight by regulatory authorities. Passengers, government regulators, and bus operators form the key user personas affected by this deficiency.

A second challenge is the **inequity of the fixed annual permit fee structure**. Bus owners are charged a predetermined amount regardless of the actual distance travelled. This creates financial strain, particularly for small-scale operators when buses experience breakdowns or reduced operational days. This is a key issue for private sector bus operators who mainly focus on profitability.

A third issue is the **lack of bus availability**, especially after midnight, when profit-seeking bus owners and drivers are discouraged by low passenger numbers and insufficient financial incentives. This affects financially vulnerable groups who depend on late-hour mobility, including shift workers, emergency travelers, and students.

These problems collectively create social, economic, ethical, and technical impacts such as reduced accessibility, unfair financial burdens, inefficient regulatory enforcement, and unsafe nighttime travel conditions. They also pose institutional risks for the government, which requires verifiable, data-driven regulatory frameworks.

Addressing this problem contributes directly to several Sustainable Development Goals like SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 8 (Decent Work and Economic Growth), and SDG 16 (Peace, Justice, and Strong Institutions). A well-designed and developed smart, transparent, GPS-driven management system will enable fair regulation, improved accessibility, and safer, more reliable nationwide mobility.

References

[1] M. G. M. M. M. A. a. M. A. A. Al-Fuqaha, " Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials,* p. 2347–2376, 2015.

[2] A. H. a. A. M. M. A. Hannan, "A GPS-Based Vehicle Tracking System with Real-Time Monitoring," *International Journal of Computer Science and Network Security,* vol. 12, p. 1–8, 2012.

[3] S. K. a. P. K. Jain, "Smart Transportation Systems Using Internet of Things," *International Journal of Advanced Research in Computer Science,* vol. 7, pp. 34-39, 2016.

[4] R. K. K. a. S. Yerroju, "Smart Traffic Management System Using IoT," *IEEE International Conference on Computational Intelligence and Communication Technology (CICT),* pp. 1-6, 2018.

[5] M. G. K. a. J. C. G. E. I. Vlahogianni, "Short-Term Traffic Forecasting: Where We Are and Where We're Going," *Transportation Research Part C,* vol. 43, pp. 3-19, 2014.

Commented [BA1]: Mention citations in the paragraphs where you have the details taken from research papers

6. Brief description of Existing Research and Systems (300 words max)

A wide range of intelligent transport solutions already exist globally, focusing on GPS-based tracking, schedule prediction, and fleet management. Commercial systems such as Google Transit**,** Moovit, and NextBus provide real-time bus arrival information using GPS feeds from transport authorities. In countries like Singapore and the UK, government-operated systems track route adherence and vehicle performance using automated vehicle location (AVL) technology. These products demonstrate the feasibility of real-time tracking but are designed primarily for passenger information.

Academic research has further advanced in route modeling and GPS-based fleet analytics. Studies on Real-Time passenger Information Systems introduce algorithms for route creation, link-based travel time estimation, and real-time deviation detection. Also, GPS and Wi-Fi-based PTMS frameworks explore low-cost data collection, on-board storage, and periodic syncing. IoT-driven fleet management research also demonstrates trip segmentation and predictive maintenance. These works collectively provide scalable methods for tracking bus movement, detecting anomalies, and estimating distances travelled.

Despite these advancements, several **critical gaps** remain, especially for use cases relevant to Sri Lanka. First, existing systems **rarely integrate regulatory enforcement**, meaning they cannot verify whether buses are operating strictly within their legally approved routes. For example, even though India's BMTC uses GPS tracking for passenger updates, buses can still deviate from routes without automatically triggering regulatory actions. Second, there is **no established framework** linking real-time distance travelled to government permit fees. Current fee models in many developing countries remain fixed and disconnected from actual bus usage. Third, existing solutions do not address **24/7 service** through data-driven incentive mechanisms.

Therefore, current research and commercial systems provide strong technical foundations but do not solve the combined problem of route compliance, distance-based permit fee calculation, and incentivized low-demand operations**,** which your project aims to address.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Existing systems mainly provide passenger-facing information or basic fleet tracking, but they do not integrate regulatory enforcement, distance-linked permit billing, or strategies to ensure 24/7 driver incentives. This proposed system bridges these gaps by introducing the following components:

1) Junction-to-junction route enforcement

From each major junction, a GPS tracker sends a message to the system containing the bus's exact coordinates and timestamp. The backend maps these events to the legally approved sequence of junctions on the permit route. Any missed, skipped, reversed, or extra junction triggers a "route deviation alert" for regulators. This provides verifiable digital evidence, something current operator-led GPS systems do not offer.

2) Accurate distance-based permit calculation

Using validated junction-sequence data, the system reconstructs each completed trip and calculates the total distance in kilometers. Permit fees are computed based on:
Permit Fee = Distance × Rate per km

This replaces the unfair fixed annual fees with a transparent, usage-based model.

3) Off-peak service incentives

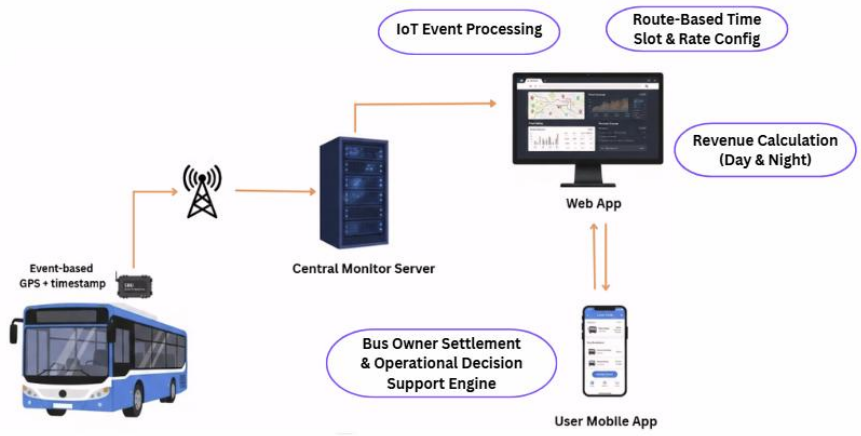
The system automatically tags trips completed during low-demand hours (10 PM–5 AM). The bus owner is awarded a higher per-km commission for these periods. This motivates operators to maintain late-night services, addressing a major gap in current practice, and to offer 24/7 services.

4) Driver and Owner empowerment through a mobile app

The app displays live earnings, trip history, breakdown submission, night shift bonuses, notifications, and other analytics and predictions. Through the app, Drivers can also switch to **"Personal Trip Mode"**, which temporarily disables regulatory GPS tracking during non-business and personal trips.

System perspective of the solution in the form of Input, Process, and Output:

- Input:
  - GPS junction messages (busID, junctionID, coordinates, timestamp)
  - Permit route geometry (ordered junctions)
  - Driver flags (breakdowns, personal trip mode on/off)
  - Policy parameters (rates, time buckets, penalties, incentives)
- Process:
  - Map matching and junction-sequence validation
  - Trip segmentation and distance reconstruction
  - Route deviation detection and alert generation
  - Billing engine: distance-based permit fee + breakdown deductions
  - Incentive engine: off-peak commission calculation
  - Data storage, analytics dashboards, and secure audit logs
- Output:
  - Real-time regulator dashboard with route compliance status
  - Daily distance logs and weekly/monthly permit fee reports
  - Driver commission summaries and earnings breakdown
  - Several analytics and predictions, such as new permit fees



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

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- **Gaining Specialized Domain Expertise**

Members can gain domain expertise primarily by engaging with Sri Lankan transport authorities, such as the National Transport Commission (NTC), Provincial Road Passenger Transport Authorities (PRPTAs), and private bus operator unions. These stakeholders can provide insights into current permit policies, route permit structures, common challenges in route enforcement, and operational pain points. Reviewing official documents such as government transportation acts, fare regulations, and bus permit guidelines will further enhance conceptual understanding. Additionally, domain expertise can be strengthened through the study of existing research papers and systems.

- **Gaining Technical Knowledge and Skill**

Technical expertise such as GPS integration, mobile app development, backend systems, and map matching algorithms can be acquired through project-based learning, modern online resources, and proper communication with colleagues who have expertise in different domains.

- **Data Required and How It Is Collected**

Data required for this project includes:

- Bus GPS coordinates and timestamps
- Permit route geometry
- Driver activity data
- Distance logs and trip reconstructions

Data can be collected using prototype GPS devices mounted temporarily on selected buses or via controlled field tests on private vehicles running along designated routes. Essential data, like sample routes or historical logs, can also be collected from government organizations such as the National Transport Commission (NTC) and Provincial Road Passenger Transport Authorities (PRPTAs) to further validate algorithms.

- **Ethical Clearance Requirements**

Because the system involves tracking vehicle movement and collecting driver behavioral data, ethical considerations are significant. To ensure ethical compliance, the project team must have informed consent from drivers and bus owners while also restricting access to sensitive data and logs. If testing is conducted only on personal vehicles or synthetic data, formal ethical clearance may be minimal or not required.

9. Objectives and Novelty

Main Objective To develop an IoT-enabled, event-based Smart Bus Route Monitoring and Revenue Management System that automates time-slot-based revenue calculation and night-time incentive distribution for public transport operations within the Western Province of Sri Lanka.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Ahamed M.T.M IT22121042	To develop an event-driven processing engine that updates bus location and trip progress using GPS and timestamp data received from on-board IoT devices.	• Set up the GPS device inside buses to capture coordinates and timestamps. • Implement event-based data transmission for start and arrival times, route deviations, and halt detection. • Implement map-matching techniques using official bus route polylines. • Detect halt-to-halt movement events based on geo-fencing logic. • Implement tamper detection and offline status reporting.	This research presents a novel event-based IoT tracking solution optimized for Sri Lankan public buses, utilizing Wi-Fi–based data transmission for efficient communication under low operational costs. The proposed system transmits only significant operational events, reducing bandwidth usage while ensuring reliable delivery of route compliance and status data in real-world public transport conditions.

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		Validate device reliability and data accuracy through real-world field testing.	
Gunasekara W.S.H.R IT22228208	To design a configurable policy engine that manages time-slot definitions and route-specific rate structures for bus operations within the Western Province.	• Define configurable time slots, including peak hours, school hours, night service, weekends, and public holidays. • Split trip distances across time-slot boundaries using IoT event timestamps. • Identify incomplete routes and missed mandatory peak-time services. • Enable manual decision-making in times of need (e.g., Bus breakdowns).	This component introduces a policy-aligned, time-slot-based rate configuration framework that complements the existing permit system by enabling operational usage-based calculations.
Sachintha H.D.D IT22264152	To implement a calculation engine that determines government revenue from daytime operations and incentive payments to bus owners for nighttime services.	• Calculate payable revenue using a distance × time-slot multiplier model. • Computer peak and normal time revenues are payable to the government. • Calculate night-time incentives payable to bus operators. • Generate daily and monthly revenue and incentive summaries.	This research introduces a hybrid revenue–incentive calculation model that transitions from annual fixed permit fees to a usage-based, time-sensitive system.

Aththanayaka N.D IT22925268	To develop a mobile-based operational and settlement support component that enables bus owners to manage registrations, review monthly financial obligations, and support manual decision-making using time-slot-based operational data.	• Develop a digital permit registration and management module for bus owners and operators. • Show the amount of daily earnings by running night shifts. • Deliver notifications to bus owners via mobile app. • Maintain permit status history linked to buses and operators. • Support manual annotations and decisions (e.g., breakdowns, emergencies).	This component introduces a data-driven operational transparency platform for bus owners, enabling informed decision-making and financial clarity without relying on manual reporting or enforcement-heavy monitoring mechanisms.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	√	No	

b) The proposed topic exhibits novelty.

Yes	√	No	

c) The student group can successfully execute the proposed project.

Yes	√	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	√	No	

e) Any other comments:

It is not NTC WebApp, it should be the Road Passenger Transport Authority - Western Province

	Title	First Name	Last Name	Signature
Supervisor	Mr	Sanjeeva	Perera	
Co-Supervisor	Ms	Buddhima	Attanayaka	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				